

THOMAS POINT, MD, USA

WMO: 994400

Lat: 38.899N

Lon: 76.436W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	21.1	-3.7	4.5	21.8	1.3	5.9	23.6	40.3	43.8	35.5	37.7	17.1	310	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	87.0	76.4	85.0	75.9	83.3	75.2	81.8	83.4	79.9	82.0	78.2	80.8	10.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.4	82.5	79.3	151.9	81.0	77.5	142.8	79.6	45.5	83.2	43.3	81.9	41.6	80.7	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.1	31.9	26.6	DB	13.5	91.4	5.9	2.5	9.2	93.2	5.8	94.7	2.5	96.1	-1.8	98.0	(4)
(5)			WB	10.7	83.3	4.0	3.4	7.9	85.8	5.6	87.8	3.3	89.8	0.4	92.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	35.5	37.3	44.2	55.0	64.2	72.8	77.5	76.6	70.6	60.0	49.5	40.7
	DBStd	16.21	9.01	7.85	7.67	6.91	6.26	5.28	3.89	3.81	5.05	6.66	7.23	7.60
	HDD50	1457	454	359	212	27	0	0	0	0	0	7	95	303
	HDD65	4153	914	774	644	308	93	8	0	0	11	183	464	754
	CDD50	4050	6	5	33	177	441	683	852	824	619	315	81	14
	CDD65	1271	0	0	0	8	69	241	387	359	180	27	0	0
	CDH74	7765	0	0	3	28	191	1236	3037	2541	680	49	0	0
	CDH80	1384	0	0	0	1	22	185	667	449	53	7	0	0
Wind	WSAvg	13.1	13.7	13.7	14.9	14.4	13.0	11.8	11.2	11.3	12.6	13.4	13.8	13.3
Precipitation	PrecAvg	46.0	3.5	2.6	4.3	3.6	3.7	4.2	4.4	4.5	4.3	3.9	3.1	3.7
	PrecMax	65.9	6.9	7.7	9.3	6.6	8.2	12.7	8.5	8.1	12.3	8.2	6.5	8.2
	PrecMin	35.3	1.4	0.3	0.0	1.5	0.7	2.2	1.6	1.2	0.6	0.1	0.6	1.4
	PrecStd	6.9	1.3	1.7	1.9	1.2	2.0	2.4	2.0	1.9	2.7	2.4	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	63.4	70.3	78.0	83.2	87.6	91.2	89.2	84.7	78.4	68.7	64.5
		MCWB	54.9	53.4	60.4	62.3	73.8	74.3	78.7	74.4	74.0	72.0	61.1	58.4
	2%	DB	54.7	55.0	63.3	72.5	78.4	84.4	87.6	86.1	81.4	74.7	64.4	58.6
		MCWB	50.0	47.7	56.0	61.3	71.1	74.3	78.0	75.3	74.4	69.5	59.3	55.3
	5%	DB	50.9	50.6	58.6	68.1	75.8	82.2	85.6	84.2	79.5	72.1	61.9	54.4
		MCWB	47.7	45.1	53.1	60.1	70.2	73.9	77.3	75.0	73.3	67.5	57.8	51.6
	10%	DB	47.3	47.2	54.9	64.7	73.2	80.0	83.7	82.4	77.8	69.4	59.5	50.6
		MCWB	43.6	42.7	50.2	58.1	68.4	73.5	76.2	74.4	71.5	65.0	55.5	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	56.2	63.0	68.9	79.4	82.1	85.9	81.6	80.2	74.3	64.8	59.8
		MCDB	60.9	60.8	66.3	70.9	80.9	83.4	87.4	83.2	80.9	75.4	66.1	60.9
	2%	WB	52.2	49.5	58.4	64.9	75.0	79.3	83.1	79.8	77.6	72.1	61.2	55.1
		MCDB	54.7	52.3	61.7	69.2	76.4	80.6	84.4	82.1	79.1	74.6	63.1	56.9
	5%	WB	48.4	45.5	54.6	62.7	72.4	77.2	81.1	78.2	75.9	69.6	59.0	51.7
		MCDB	50.6	48.3	57.5	67.3	74.7	79.5	83.1	81.2	78.3	71.8	60.7	53.3
	10%	WB	44.3	42.2	51.2	60.3	70.4	75.5	79.0	76.9	74.3	66.8	56.5	48.3
		MCDB	46.4	45.1	53.7	64.4	73.0	78.6	81.8	80.3	77.0	69.3	58.3	49.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	11.1	12.0	10.4	9.2	8.9	8.5	8.6	9.4	9.4	8.9
		MCDBR	14.9	15.8	17.2	17.1	12.8	11.1	10.6	9.9	9.2	11.0	11.8	13.6
		MCWBR	12.7	12.8	12.1	9.4	7.9	8.0	7.9	7.3	7.3	8.6	10.2	13.0
	5% WB	MCDBR	15.5	14.4	16.4	14.9	10.5	9.8	9.2	8.6	8.1	9.7	10.1	12.7
		MCWBR	13.9	12.7	13.6	10.5	7.5	7.9	8.2	7.5	7.4	8.9	10.2	13.2
Clear-Sky Solar Irradiance	taub	0.312	0.320	0.359	0.398	0.442	0.494	0.518	0.511	0.421	0.362	0.326	0.311	
	taud	2.484	2.446	2.394	2.304	2.214	2.114	2.062	2.067	2.291	2.461	2.519	2.526	
	Ebn at Noon	274	287	285	279	267	253	245	243	261	268	267	267	
	Edh at Noon	25	30	35	40	45	50	52	50	38	29	24	23	
All-Sky Solar Radiation	RadAvg	667	935	1243	1584	1787	1976	1942	1726	1402	1048	758	586	
	RadStd	52	103	99	121	212	150	93	98	154	114	65	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	N/A
(47) Regional (81 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+148

Nomenclature: See separate page